

Deepak David Joseph Periyamayagam

Lead UI Engineer & Web Architect

12 Auldfield Lane, Glasgow, United Kingdom, +447497355038, dpkdavid@yahoo.com

LINKS

[LinkedIn](#), [Github](#), [Website](#)

PROFILE

Engineering leader and Web Architect with 11 years in UK financial services (JP Morgan Chase, Barclays), leading the architecture and delivery of secure, scalable web platforms for regulated environments. Owns end-to-end web application architecture — React, Next.js, TypeScript, design systems, and micro-frontends — backed by CI/CD on Docker and Kubernetes, with measurable outcomes: 80% reduction in end-user effort, 50% uplift in user satisfaction (CSAT), and time-to-production cut from 60 to 20 minutes. Drives AI adoption across engineering, pioneering LLM-driven Angular-to-React conversion and OpenAPI-aware declarative UI generation. Builds high-performing teams through structured mentoring, architecture governance, and engineering standards spanning WCAG 2.2 AA accessibility, mutation testing, and observability.

EMPLOYMENT HISTORY

Mar 2024 — Present

Lead UI Engineer, JP Morgan Chase

Glasgow, UK

Compliance Central Dashboard

Monorepo, Yarn Workspaces, React, TypeScript, Vite, Storybook, Axe, Turborepo, Vitest, RTL, Cypress, Stryker, Docker, Kubernetes, Google Analytics, SonarQube, Jira, REST API Gateway, Dynatrace, Teams

- Built a scalable portal hosting multiple compliance apps and dashboards for compliance officers; cut end-user effort by **80%** via guided workflows and streamlined monitoring.
- Added end-to-end auditability (immutable event logs, trace IDs, Dynatrace) to satisfy audit and speed investigations.
- Integrated **Web Speech API** for voice-to-text; raised user satisfaction by **50%** (CSAT).
- Standardized a11y checks (**WCAG 2.2 AA**), linting, and Storybook docs; eliminated duplicate code and circular dependencies.
- Improved performance with **code splitting, lazy loading**, and prefetch; enforced Lighthouse and bundle-analysis gates pre-merge.
- Raised release confidence with Vitest/RTL + Cypress and **Turborepo caching**; reduced time-to-production from **60 min** → **20 min**, including security and **Sonar** scans.
- Implemented **mutation testing with Stryker** to harden test effectiveness beyond line coverage; achieved a **[X]% mutation score** and eliminated weak tests before release.
- Deployed on **Kubernetes** with environment-aware configurations and health checks for safe rollouts.

Compliance Component Library

React, TypeScript, Storybook, Vite/Rollup, Vitest; Jira, Confluence

- Created a reusable, tree-shakeable component library with built-in a11y patterns; reduced duplication and time-to-first-feature across multiple teams.
- Enabled **tree shaking** to minimize shipped JS.

Compliance Widgets (Microfrontends)

Vite Module Federation, React, TypeScript, Vitest, Storybook; Jira

- Delivered embeddable, federated widgets enabling incremental adoption and **independent deployments**.

AI Enablement

Prompt engineering, context engineering, shot prompting, AI agent skills, AI review agents

- Introduced AI-assisted code and test generation; halved developer effort and cut PR lead time.
- Built AI agents that review junior developers' code on their machines pre-PR, scoring **security, performance, accessibility, complexity, and code quality**; PRs carry the pass/fail verdict, screenshot evidence, and justification, cutting review effort by **[X]%**.
- Designed and rolled out **AI agent skills** for the Compliance Central Dashboard, automating recurring engineering and compliance workflows; cut manual effort by **[X]%**.

Compliance Ecosystem (Director View)

React, TypeScript, Next.js, React Flow, Framer Motion; Jira

- Built an interactive map of systems, data flows, dependencies, and accountable owners.
- De-risked program decisions and accelerated cross-portfolio planning.

ARMigrator — AI-Driven Angular → React Conversion

TypeScript, Claude Sonnet 4, curated knowledge base, prompt pipelines; Jira, Confluence

- Led the **winning hackathon prototype** and championed it into an active migration programme.
- Built an LLM conversion pipeline (curated knowledge base, automated validation) converting legacy **Angular** to production-grade **React**; cut migration effort per module by **[X]%**.

Declarative UI Generation — OpenAPI-Aware LLMs (R&D)

React, TypeScript, OpenAPI, JSON Schema, LLMs, design systems

- Conceived and prototyping a **declarative UI generation** approach in which **OpenAPI-aware LLMs** generate schema-driven, design-system-compliant React UIs directly from API contracts.
- Targets a **[X]%** build-time cut with a11y and design standards enforced by construction.

Web Architecture, Leadership & Mentoring

- Own the **end-to-end architecture** of web applications across the portfolio — technology selection, design reviews, non-functional requirements, and integration patterns.
- Mentor **3 junior web developers** on best practices (code quality, testing, accessibility, performance) via code reviews and pairing; coach **1 engineer** on career growth.
- Partner with product owners and directors to shape roadmaps and architecture trade-offs.

Nov 2022 — Mar 2024

Assistant Vice President, Barclays

Glasgow, UK

Developer Knowledge Hub

React, TypeScript, Docusaurus, Vite, Jest; Jira, Confluence, Teams

- Built a documentation platform consolidating install guides, runbooks, and tooling standards; cut Confluence sprawl and enabled deprecation of legacy toolchains to reduce license costs.

Mar 2017 — Oct 2022

Senior Associate, JP Morgan Chase

Glasgow, UK

React, TypeScript, Redux, Highcharts, WebSockets; Jest, React Testing Library, Cypress; Cloud Foundry, Jenkins; Jira, Confluence

- Built high-throughput post-trade dashboards over REST microservices and **WebSocket** streams; achieved **~85%** coverage with layered unit/integration/E2E tests.
- Operationalized **CI/CD to Cloud Foundry** with reusable Jenkins pattern builds and blue-green deploys; increased deploy frequency and cut change-failure rate by **~90%**.
- Led and mentored a **four-engineer squad**; standardized an internal React toolkit (a11y, linting, Storybook) that removed duplicate UI code across teams.

Sep 2015 — Feb 2017

Associate, JP Morgan Chase

Glasgow, UK

Spring Boot, AngularJS, Java, Hibernate, REST; JUnit, Cucumber, CI; Jira

- Built a workflow automation platform for margin-call processing on resilient REST services; optimized queries and caching for low latency at scale.
- Implemented **BDD** quality gates (JUnit, Cucumber) in CI; reached **~80%** coverage and materially reduced regressions.

EDUCATION

Sep 2012 — Sep 2013	Master of Science in Microelectronics System Design, University of Southampton	Southampton, UK
	Graduated with Distinction	
Jun 2006 — Jun 2010	Bachelor of Engineering in Electronics and Communication, Karunya University	Coimbatore, India
	Graduated with Distinction	

SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, React Flow, TanStack React Query

Quality & Testing: Vitest, Jest, React Testing Library, Cypress, Stryker (mutation testing), Storybook visual testing, Axe, WCAG 2.2 AA accessibility

Platform & Tooling: Docker, Kubernetes, CI/CD, Turborepo, Yarn workspaces, Vite, Rollup, NPM, REST, WebSockets, OpenAPI

AI Engineering: LLM integration, prompt & context engineering, AI agent skills, AI-assisted migration and code generation

Leadership: Web architecture, technical leadership, mentoring & coaching, senior stakeholder communication

LANGUAGES	English	Highly proficient	Tamil	Native speaker
-----------	---------	-------------------	-------	----------------